Scholar No:..... Scholar Name:.....

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
END TERM EXAMINATION (December 2024)

Course: M.Tech. Semester: I Date:21/12/2024
Time: 2 hour Max. Marks: 40

Sub: Web Search and Information Retrieval

Branch: All Branches Sub Code: FAC-560/FCN 558/ FIS 561/ FAI 558

Note: Attempt All Questions.

Q. No.	Questions	Marks /CO's
1.	Given the following set of documents, construct an inverted index and compute the following for the term "retrieval": I. **Collection frequency (CF)**. II. List of documents containing the term along with **term frequency (TF)** in each document. III. **Positional information** for the term in the documents. IV. Using **proximity operators**: a. Identify in which document the term **"retrieval" OW/3 "information"** appears. b. Identify in which document the term **"retrieval" UW/2 "information"** appears. **Note:** Perform only **tokenization** as preprocessing. Do not apply lemmatization or stop word removal. ***Documents:*** ***Doc 1:*** "Information retrieval provides powerful results." ***Doc 2:*** "Information systems rely on retrieval and indexing." ***Doc 3:*** "An inverted index aids information for retrieval."	05 [CO1]
2.	Consider the Boolean query **mercy AND worser**. The postings list of the two words are shown. The list for mercy is augmented with skip pointers: **mercy → [4, 6, 10, 16, 25, 40, 61, 69, 76, 82, 85, 90, 93, 98, 101, 116]** **worser → [4, 6, 90, 96, 100, 101, 102]** Answer the following questions: I. How many comparisons are required if neither of the lists had skip pointers? List the comparisons sequentially. II. How many comparisons are required when Mercy has skips? List the comparisons sequentially. III. What is a common heuristic for deciding the number of skip pointers, given that the size of the postings list is P?	05 [CO2]
3.	A search engine indexes a collection of **1 million documents**, each with an average of **500 words**. To efficiently build and store the index, the following indexing and compression techniques are considered: assum t = 1 i. Assume the term frequency follows **Zipf's Law**, where the frequency of a term is inversely proportional to its rank. If the most frequent term appears **10 million** times, estimate how many times the **2nd, 10th, and 100th** ranked terms will occur. ii. Given that the total number of unique terms V in a collection is estimated by **Heap's Law** ($V=K.N^b$). Where N is the total number of terms, k =30, and b =0.5.	05 [CO3]

	Estimate the size of the vocabulary for the given collection (1 million documents, average 500 words). iii. Explain how the **Blocked Sort-Based Indexing** method works for this dataset. Given that the system memory can only hold **100,000** term occurrences at a time, calculate the number of blocks needed to process the collection in a single pass.	
4	The system maintains an inverted index and utilizes **B-Trees and Reverse B-Trees** for efficient wildcard query processing. Consider the following two wildcard queries: **Query 1:** Jayasur * a **Query 2:** Sana* AND Jayasur* Describe how you would process the wildcard query. Additionally, explain the difference in processing.	05 [CO3]
5.	The Soundex algorithm is used to encode words into phonetic representations based on their pronunciation. Consider the following list of names: **Catherine, Miller, Kathryn, Malar** i. Calculate the Soundex code for each name in the list. ii. Find the Soundex matches for the following names: **Query 1:** Kathleen **Query 2:** Mohler	05 [CO4]
6.	Consider the task of spelling correction for an isolated word using the Levenshtein Edit Distance algorithm. The word "**LEVENSHTEIN**" has been misspelled as "**MEILENSTEIN**". Determine the minimal set of edit operations (insertions, deletions, substitutions) required to correct the misspelling.	05 [CO4]
7.	Given the following corpus of documents and a query term "data", calculate the **Term Frequency-Inverse Document Frequency (TF-IDF)** score for the term "data" in each document. Corpus: N=3 **Document 1:** "Data science is a field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from structured and unstructured data." **Document 2:** "Data analysis involves inspecting, cleaning, transforming, and modeling data to discover useful information." **Document 3:** "Big data technologies enable the analysis of large and complex data sets."	05 [CO5]
8.	Given the following term frequency table for a set of documents and a query term. Assume that the log frequency is 1 for all terms, and normalization must be performed on the term frequency vector (using Euclidean normalization). The term frequency table is as follows: i. Calculate the cosine similarity between the query and each document. ii. Determine the order of document relevancy based on the rank.	05 [CO5]

Term	Doc 1	Doc2	Doc 3	Query
Data	3	2	1	2
Science	2	0	0	1
Analysis	1	2	0	0
Knowledge	0	1	2	1
Algorithms	1	0	1	0
Systems	0	1	0	1

***Subject Coordinator: Dr. Muktesh Gupta**

Scholar No: Ruhun Dwivedi Scholar Name: 242120 11118

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
MIDTERM EXAMINATION (November 2024)

Course: M.Tech. **Semester: I** **Date:23/11/2024**
Sub: Web Search and Information Retrieval **Max. Marks: 20**
Time: 1 hour
Branch: All Branches **Sub Code: FAC-560/FCN 558/ FIS 561/ FAI 558**

Note 1: Attempt All Questions.

Q.No.	Questions	Marks	Cos
1.	Consider the following documents: **Doc1:** "The skilled engineer designs a robust system." **Doc2:** "A talented programmer writes efficient code." **Doc3:** "The experienced manager leads a productive team." a) Construct the inverted index. Assume stop words (e.g., "the," "a," "an") are removed, and all terms are converted to lowercase. b) Determine which documents are retrieved for the following queries: **Query 1:** "engineer AND system" **Query 2:** "team AND NOT productive"	05	CO1
2.	Discuss the importance of evaluating an information retrieval system and describe the key performance metrics used for the evaluation. Two information retrieval systems, IRX and IRY, are tested on the same query over a collection of 500,000 documents. IRX retrieved 40,000 documents, out of which 6,000 are relevant, while IRY retrieved 50,000 documents, out of which 4,000 are relevant. Assuming the total number of relevant documents in the collection is 10,000: a) Compute **Precision** and **Recall** for both systems. b) Identify which system performs better based on these metrics and justify your conclusion.	05	CO2
3.	Consider the following set of words in a vocabulary: Words: teacher, doctor, engineer, artist, scientist a) Construct the permuterm index for the given words. b) Using the constructed permuterm index, determine which words would be retrieved for the following queries: **Query 1:** *ist **Query 2:** eng* **Query 3:** t*er	05	CO2
4.	The Soundex algorithm is used to encode words into phonetic representations based on their pronunciation. Consider the following list of names: **Robert, Rupert, Ashcraft, Pfister, Honeyman** a) Calculate the Soundex code for each name in the list. b) Find the Soundex matches for the following names: **Query 1:** Rupert **Query 2:** Ashcroft	05	CO3